

# American Ev - Data Analysis Report

## Project Overview

This project analyzes **Electric Vehicle (EV) population data** to understand adoption trends, geographic distribution, vehicle characteristics, and market growth patterns. The analysis is designed to support insights into how EV adoption is evolving across regions and over time.

The project was developed using **Tableau** for visualization and a structured **CSV dataset** as the data source.

## Dataset Description

### Electric\_Vehicle\_Population\_Data.csv

This dataset contains detailed information about registered electric vehicles, including: - Vehicle Make and Model - Model Year - Electric Vehicle Type (BEV / PHEV) - Geographic attributes (State, County, City, ZIP) - Electric range and eligibility indicators

The dataset enables **time-based**, **geographic**, and **product-level** analysis of EV adoption.

## Data Preparation & Modeling

- Data types were validated (dates, numeric ranges, categorical fields)
- Null and inconsistent values were reviewed
- Geographic fields were standardized for mapping
- Measures and dimensions were prepared for Tableau analysis

This ensured accurate aggregation and filtering across dashboards.

## Electric Vehicle Adoption Trends

### Growth Over Time

- EV registrations show a **strong upward trend**, especially in recent years.
- Adoption accelerates significantly after major technological improvements and policy incentives.

**Insight:** EV adoption is no longer linear; it demonstrates exponential growth patterns.

## Geographic Distribution of EVs

### EV Population by Region

- Certain state Like Washington dominates EV adoption.

**Insight:** Infrastructure availability and income levels strongly influence EV adoption.

## Manufacturer & Model Analysis

### Top EV Manufacturers

- A small number of manufacturers account for a large share of EV registrations.
- Market leaders benefit from early entry and extensive charging ecosystems.

### Popular EV Models

- Tesla models consistently rank highest in registrations.
- Consumer preference favors models with higher range and reliability.

**Insight:** Brand trust and range anxiety are major decision factors.

## Electric Vehicle Type Analysis

### BEV vs PHEV

- Battery Electric Vehicles (BEVs) significantly outnumber Plug-in Hybrid Electric Vehicles (PHEVs).
- BEV adoption increases as charging infrastructure expands.

**Insight:** Consumers are shifting toward fully electric solutions rather than transitional hybrids.

## Range & Technology Insights

- Vehicles with higher electric range show greater adoption.
- Improvements in battery technology directly correlate with increased registrations.

**Insight:** Technological advancement is a key driver of EV market growth.

## Business & Policy Recommendations

1. **Expand Charging Infrastructure:** Focus on underrepresented regions to balance adoption.
2. **Promote Long-Range Models:** Address consumer range anxiety.
3. **Target Urban Growth Markets:** Cities show the fastest adoption rates.
4. **Policy Incentives:** Tax credits and rebates strongly support adoption growth.
5. **Manufacturer Strategy:** Innovation and affordability are critical for market leadership.

## Conclusion

This analysis highlights the rapid growth and evolving dynamics of the electric vehicle market. By combining geographic, temporal, and product-level insights, the project demonstrates how data analytics can support strategic decisions in sustainability, transportation planning, and automotive innovation.

The project showcases strong skills in: - Data cleaning and preparation - Exploratory data analysis - Interactive dashboard creation in Tableau - Insight-driven storytelling

This analysis is well-suited for inclusion in a **data analytics portfolio or professional presentation**.

**Tools Used:** Tableau, CSV Dataset